

Jiayu (Jerry) Liu

Incoming Undergraduate, Stanford University (Fall 2026)
 Mathematics — Algorithms — AI Research
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Education

Stanford University - Incoming Math and CS Undergraduate (Fall 2026)	2026.09 – 2030.05
The Peddie School (Valedictorian)	2022.09 – 2026.05
• Advanced Coursework: Multivariable Calculus, Linear Algebra, Algorithms & Data Structures • University Coursework (Princeton): Real Analysis (A), Fourier Analysis (A), Numerical Analysis (A), Probability and Stochastic Systems (A)	
Research Science Institute (RSI)	2025.07 – 2025.08
Ross Mathematics Program	2024.07 – 2024.08

Research Experience

Massachusetts Institute of Technology – Seethapathi Lab	2025.07 – 2025.08
<i>Research Intern</i>	
• Developed the transformer-based pipeline for 3D human pose dynamics (PyTorch) • Paper accepted to the <i>AAAI 2026 Workshop (AI4TS: AI for Time Series)</i>	
Princeton University – Department of Mathematics	2024.11 – 2025.11
<i>Research Assistant</i>	
• Paper published in <i>Transactions of the London Mathematical Society</i> • Conducted mathematical research on configuration spaces with Fields Medalist Charles Fefferman	
MIT-PRIMES (Math Track)	2024.11 – 2026.01
<i>Research Scholar</i>	
• Developed algebraic methods for studying the fundamental groups of hyperplane arrangements • Mentored by Prof. Nathan Williams (University of Texas at Dallas)	

Selected Projects

Multimodal AI Assistive Device for Hearing Impaired	2024.06 – 2025.11
• Designed and implemented a low-cost multimodal deep learning system integrating computer vision, audio processing, and wearable hardware using Python and Pytorch • Implemented real-time sound localization, audio classification, and text-to-speech pipelines	
Topological Data Analysis for Alzheimer’s Diagnosis	2023.12 – 2024.11
• Applied persistent homology combined with machine learning for Alzheimer’s disease diagnosis • Published in the <i>National High School Journal of Science</i>; mentored by Prof. Dawid Kielak (Oxford)	

Skills

- **Languages:** Python, C++, MATLAB (*Familiar:* Java)
- **Technologies / Frameworks:** PyTorch, Pandas, NumPy (*Familiar:* Git)

Awards and Recognitions

- **Science Olympiads:** USA Math Olympiad Bronze, USA Computing Olympiad Gold Division
- **Industry Recognition:** Full-Time Exceptional Engineer Offer from xAI